FRC STRAIGHTNESS ADJUSTMENT

α0iD/1iD/0iE/1iE/C400iA/C600iA

Preferred By
K.BALAMURUGAN
Service SZ

TOOLS REQUIRED FOR DOING THIS ACTIVITY

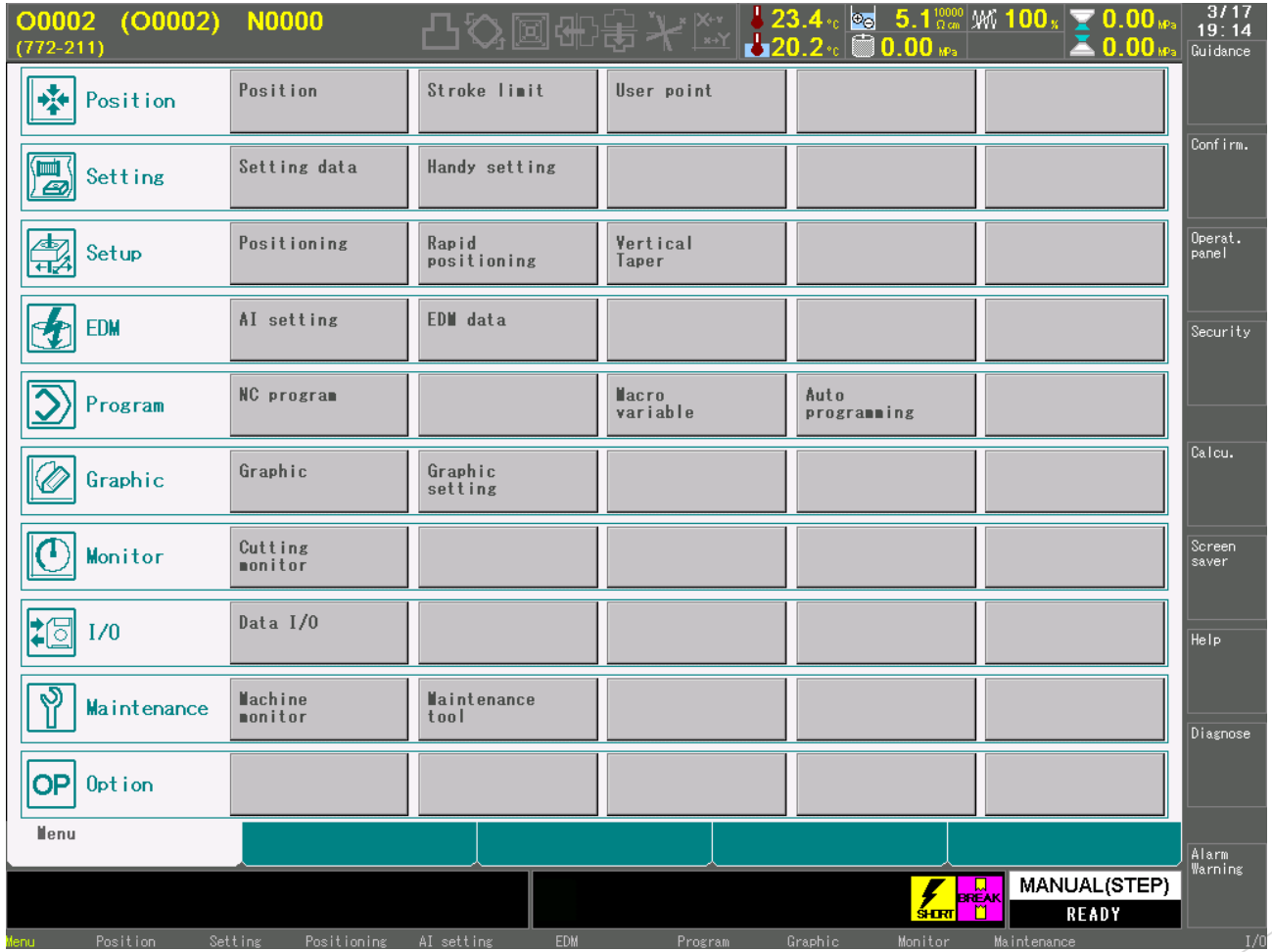
- Cylinder Master(300mm)
- One Micron lever dial gauge
- Dial stand

Preparations before starting Adjustment

- Take machine SRAM back up
- Clean the machine table with soft oil stone and clean with cotton waste.
- Avoid water droplets over the cylinder master
- Before measuring , make sure that keep relay K0#7 I set to ZERO. Enable the keep relay K0#7 is 1, set 0 turn OFF the controller power and ON
- Keep the cylinder master on machine table and leave it for 15 minutes (to match the clean room temperature)

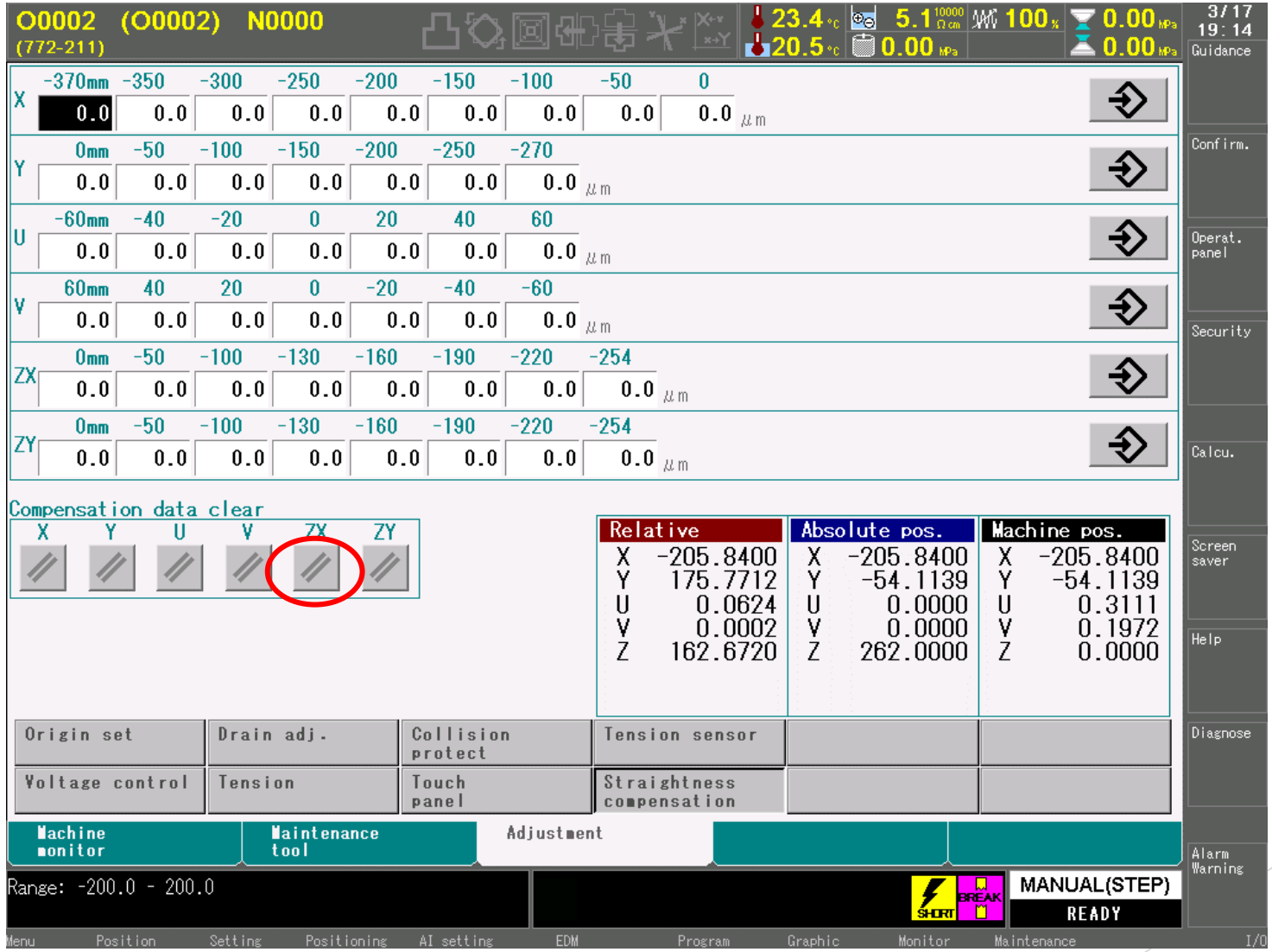
For Displaying the straightness Adjustment screen

- Press the FRC menu
- Enter 9998 and press input button in MOP , Adjustment menu option will come



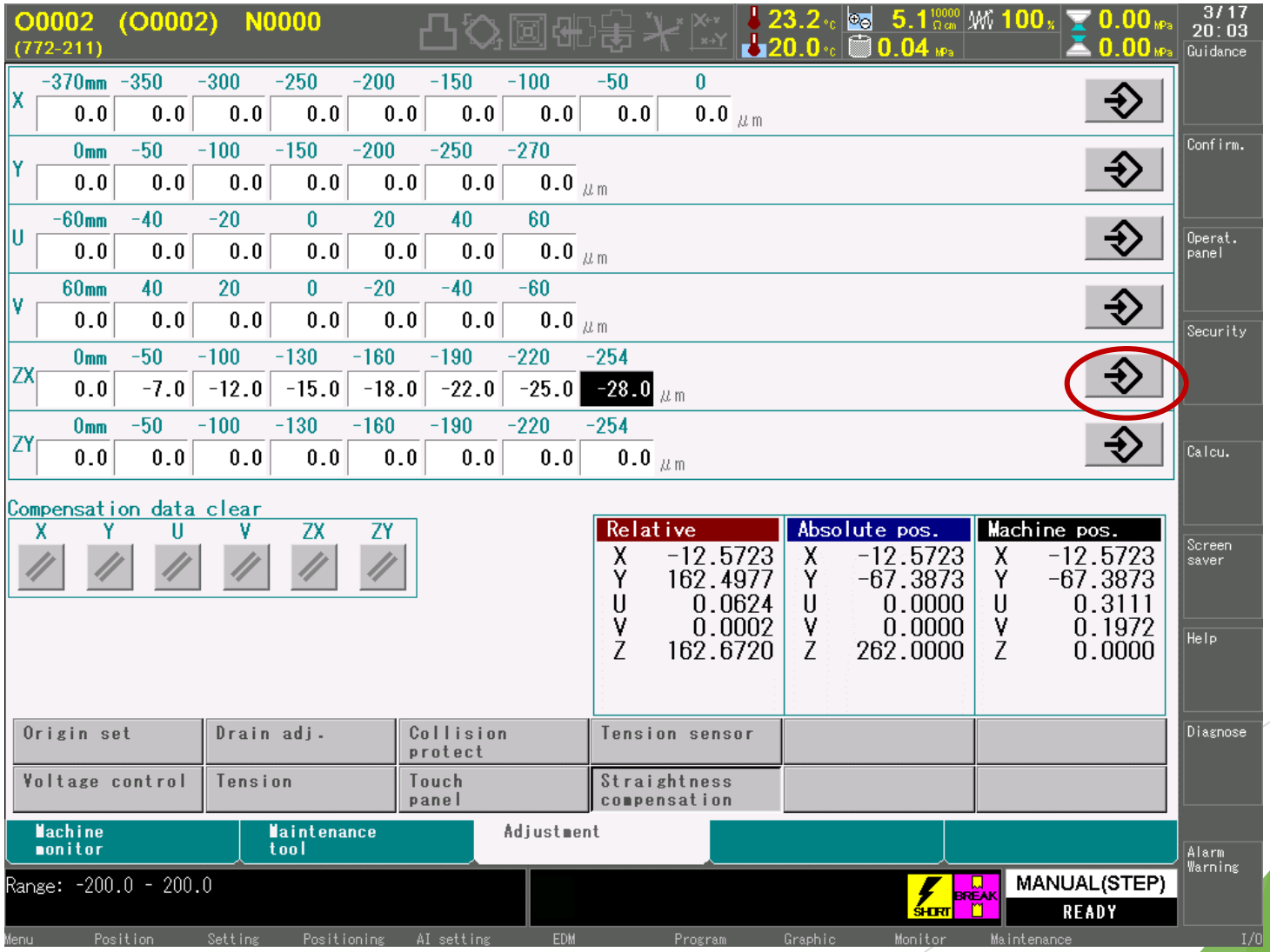
Touch the adjustment button in screen

Straightness Adjustment Screen(Press the compensation data clear)



ZX Adjustment
Move the Z axis to machine ZERO
Mount the lever dial gauge as shown below (upper guide block M8 threaded hole)
Take the high point in cylinder master before taking measurement

- Measure the error at the specified point(0.00,-50,-100,-130,-160,-190,-220,-254) of each axis using the dial gauge .Input the measured value like below screen and press the calibrate button.



ZY Adjustment –Place the dial as shown below follow the similar procedure like ZX.

1. After pressing the calibrate button this data will be renewed according to the inputted value.
2. The compensation data will be added or subtracted to the current data.
3. The straightness compensation data is stored to the pitch error compensation memory.
4. For inputting the value in screen follow the below procedure (Example)

Points	Measured Value	Compensation	second measurement	Compensation
0.00	10	-5	7	-3
-50.00	20	-10	15	-7
-100.00	30	-15	20	-9
-130.00	40	-20	30	-12
-160.00	50	-25	38	-18
-190.00	60	-30	48	-18
-220.00	70	-35	59	-19.5
-254.00	80	-40	64	-20.5

Note :After compensation when will measure the points again if the value improving little bit when you are giving compensation again never give compensation value from measured one. Need to give the compensation value from first set value either add or subtract.